PRE-IDEATHON

GENESIS IDEATHON

TEAM & DOMAIN

- Team Name – QuadriX
- Team Members – K.A.Tirunavukkarasu(CB.SC.U4AIE26067),
  Monish Karthik S(CB.SC.U4AIE26034),
  Surya P(CB.AI.U4CPS26057) ,
  Kirthi Madala (CB.SC.U4AIE26027)
- Selected Domain – Domain 1
- Problem Statement – Intelligent Land Record Digitization and Validation System



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DOMAIN PROBLEM IN OUR WORDS

Explanation Of The Problem In The Domain:

What exactly needs to be solved?

- **Fragmented land information:** RoR, deeds, EC, mutation records and cadastral/GIS data are spread across different documents and systems.
- **Difficult for citizens:** A common person cannot easily understand the older language versions, compare or verify these records and often depends on intermediaries.
- **Manual verification for officials:** Officers spend significant time comparing documents and identifying inconsistencies across records.
- **No intelligent evidence layer:** Existing digitization mainly provides access to records; it does not sufficiently extract, cross-check, explain and connect the evidence for the user.
- **Property discovery lacks trust:** Existing marketplaces focus mainly on listings, price and location based only on advertisements rather than document consistency and evidence strength.

Why is it important right now?

- India is rapidly digitizing land records, creating an opportunity to make this information more usable and intelligent.
- Property decisions involve major financial and legal risk, so users need understandable, evidence-based information before proceeding.
- **AI can bridge the gap:** Digitize old/multilingual documents, extract key fields, compare records, detect discrepancies and explain them in simple language.
- **One platform can serve both sides:** Citizens get understandable property intelligence and validation, while officers get prioritized cases and evidence-based review.
- **Future-ready property ecosystem:** The same verified property profile can support a more transparent marketplace, reducing information asymmetry and unnecessary dependence on intermediaries, or just purchase based on Ads.

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WHO IS AFFECTED?

Target users, Stakeholders, Beneficiaries

Primary users / Target group:

- **Government revenue and land-record officers** — investigate flagged properties, review evidence and take the required administrative action.
- **Citizens and property owners** — understand their property records, identify missing evidence and see unresolved issues before important property decisions.
- **Property buyers** — use evidence-backed property information for safer, more informed due diligence.

Secondary stakeholders:

- **Registration, survey and other land-record authorities** — provide or verify relevant transaction, survey and parcel evidence through authorised systems.
- **Banks, lenders and legal professionals** — use reliable, source-linked property evidence during lending or due-diligence workflows.
- **Property sellers and verified listing participants** — present an evidence-backed property profile without exposing sensitive documents publicly.

Who benefits if the problem is solved?

- **Government administration** — less fragmented case investigation, faster identification of exceptions and clearer officer workflows.
- **Citizens & buyers** — reduced information gaps, easier document understanding and clearer next steps when evidence conflicts or is missing.
- **Property ecosystem** — evidence-aware discovery and direct buyer–seller connection can reduce dependence on information intermediaries while keeping sensitive records private.

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CURRENT SITUATION & CONTEXT

Existing scenario, Background and How things work today

Existing scenario & background

- Land information is distributed across revenue/land records, registration records, survey/cadastral evidence, ECs, mutation records and supporting tax records.
- Digital systems are improving access to these records, but the information still needs to be understood and cross-checked when inconsistencies or missing evidence arise.
- Citizens and buyers face the same fragmentation when trying to understand a property's ownership, history, evidence and readiness for further due diligence.

Relevant data / facts

- A property may involve multiple evidence types such as RoR/Patta, Sale Deed, EC, mutation, FMB/cadastral evidence, tax records and other applicable records.
- The useful information is spread across different documents and sources; key fields such as owner, survey number, area, dates, transactions and encumbrances may need to be extracted and compared.
- Property discovery is often separate from detailed evidence verification, creating an information gap between a listing and the property's supporting records.

Current workflow — how things work today

1. COLLECT / ACCESS — Relevant property documents and authorised records are obtained from the applicable sources.
2. EXTRACT — Important details such as owner, survey number, area, dates, transactions, mortgage/encumbrance and other fields are read from the records.
3. VALIDATE — Relevant records are compared: Deed ↔ RoR/Patta ↔ Mutation, EC ↔ mortgage/release evidence, Deed ↔ survey/FMB/cadastral evidence, and other applicable records.
4. INVESTIGATE — If information conflicts or evidence is missing, the relevant user/officer obtains additional evidence and investigates the issue.
5. DECIDE / VERIFY — Government authorities, citizens, buyers or authorised professionals take the appropriate next step based on the available evidence; legal and administrative decisions remain with the authorised authority.
6. DISCOVER / TRANSACT — Buyers typically evaluate properties using available listing and document information, while detailed verification may happen separately before a transaction.

Current gap

- Records are becoming digital, but extraction, cross-record validation, issue identification and evidence-backed property discovery are still not presented as one connected intelligence workflow.
- This creates friction for both sides: officers need prioritised exceptions, while citizens and buyers need an understandable evidence profile before making property decisions.

KEY CHALLENGES

Main difficulties , bottlenecks , root causes and consequences.

Main difficulties & bottlenecks

- Property evidence is fragmented across deeds, RoR/Patta, EC, mutation, survey/FMB/cadastral and tax records, making one-property verification a multi-record task.
- Important information must be extracted from scans, old documents, handwritten or regional-language records before it can be compared reliably.
- Key checks — ownership, area, survey identity, transaction history, mutation, mortgages/encumbrances and missing release evidence — cannot be judged from a single document.
- Citizens and buyers struggle to understand what the records mean, what evidence is missing and whether a listed property has enough supporting evidence for further due diligence.

Root causes

- Different records are created for different purposes, stored across separate workflows and may contain different formats, names, dates, units or levels of detail.
- Document availability does not automatically create a connected property-level evidence profile; relevant records still need extraction, entity matching and cross-validation.
- Property discovery and property verification are often separate: a listing can show basic details without giving the buyer a clear picture of evidence strength, unresolved issues or missing records.

Consequences if unresolved

- Officers spend more effort locating, comparing and prioritising cases instead of focusing first on the exceptions that need attention.
- Record conflicts or missing evidence can remain unclear, increasing delays and repeated verification for citizens and authorities.
- Buyers may make decisions with incomplete understanding of ownership history, encumbrances or other property risks.
- The gap between property listings and reliable evidence increases information asymmetry and dependence on intermediaries.

EXISTING SOLUTIONS/ APPROACHES

What is already being done, and where it falls short

Existing approaches

- Land & registration systems — official land, registration and EC-related records.
- Survey / cadastral / GIS systems — parcel, boundary and measurement evidence.
- Manual due diligence — professionals compare title, mutation, encumbrance and supporting records.
- Property portals — make property discovery easier through listings and seller information.

Where they fall short

- Old, handwritten and multilingual documents still need reliable extraction before automated comparison.
- Records are not automatically resolved into one connected property/owner evidence profile.
- EC/mortgage-release, mutation, title history and conflicting measurements can require deeper cross-record reasoning.
- Buyers and citizens still lack a simple view of evidence strength, missing records and unresolved issues.
- Property discovery is often separate from evidence verification.

Open gap

No single layer connects

- **extraction → validation → missing evidence → exception prioritisation → understandable property intelligence → evidence-aware discovery.**
- The project complements official systems; it does not replace official records or authorised decisions.

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

Hardware Requirements

- Developer systems + cloud/server infrastructure for AI processing, APIs, databases and document storage.
- Optional high-quality scanners/capture devices for legacy records.

Software Requirements

- Frontend + Python/FastAPI backend.
- OCR/HTR + OpenCV + multilingual NLP/document extraction.
- AI/ML for entity matching, validation and anomaly detection.
- Database/document storage with provenance and audit trail.
- Authorised APIs/GIS integrations + authentication, RBAC and secure deployment.

Other Resources / Human Resources

- AI/ML, Document-AI, backend/frontend and cloud expertise.
- Land-record/survey/registration domain experts for validation.
- Government/field-user participation for workflow testing and human review.
- Authorised government data/API access; synthetic/representative data for the prototype.

INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

What our team intends to do

- Build the project as an AI-powered property intelligence layer that extracts, connects and validates land/property evidence.
- Support two connected sides: exception-first government validation and evidence-first citizen/buyer property discovery.

What we want to improve

- Turn complex records into comparable evidence, highlighting inconsistencies, missing evidence, encumbrances and the next verification step.
- Make property discovery evidence-aware: help buyers find properties based on their needs while showing verification status and evidence strength.

What successful improvement should achieve

- Government: faster, prioritised and auditable investigation of cases needing human attention.
- Citizens & buyers: clearer property intelligence and safer evidence-based due diligence.
- Marketplace: evidence-backed listings and direct buyer–seller discovery, reducing information asymmetry and dependence on intermediaries.

TEAM CONTRIBUTION

Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Kirthi Madala	AI / Document Intelligence	OCR/HTR, multilingual document extraction, field detection and confidence scoring
Monish Karthik . S	Validation & Land Intelligence	Cross-record validation, EC/mutation/title checks, anomaly detection and evidence scoring
K.A.Tirunavukkarasu	Full-Stack & Platform	Web portal, backend APIs, database, authentication and officer/citizen workflows
Surya P	Marketplace & Product	Evidence-first marketplace, buyer–property matching, UI/UX and user workflow testing